

NISHTHA KUKREJA

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EDUCATION

University of South Florida | B.S. Computer Science, Minor in Economics | GPA: 3.8/4.0 May 2028
Coursework: Data Structures, Probability & Statistics, Linear Algebra, Discrete Structures, Program Design

EXPERIENCE

Hexa Labs, University of South Florida | Undergraduate Research Assistant Sep 2026-Present

- Research energy-efficient AI for data centers using Spiking Neural Networks (SNNs) and Spikeformer models.
- Build Python benchmarks comparing accuracy, latency, and energy use of spiking vs. conventional networks.
- Reproduce and extend recent IEEE research with faculty and PhD researchers, prototyping AI-infrastructure tools.

Nucor | Software & Data Engineering Intern Louisville, KY | May-Aug 2026

- Built a Databricks and Streamlit predictive-maintenance platform on years of shredder-hammer data, recommending maintenance across 30+ steel yards; placed 2nd in Nucor's internal hackathon.
- Architected a five-agent AI copilot (Claude, Azure AI Foundry) that orchestrates analytics, reporting, dashboards, and recommendations across five business systems; designed for 500+ internal users, rollout pending.
- Integrated SQL- and Databricks-backed Power BI semantic models so users get chat-native dashboards and answers in 30-90 seconds instead of navigating static BI reports.
- Translated stakeholder feedback into requirements and validated copilot responses against internal test scenarios.

University of South Florida | Student Health & Wellness Center, Student Lead May 2025-Present

- Shipped 15+ production website/UI improvements (navigation, responsiveness, accessibility) for 50,000+ students.
- Promoted to Student Lead; maintained 99.7% website uptime while owning web and event delivery.

University of South Florida | Information Technology Jun 2024-May 2025

- Resolved 1,500+ issues across Microsoft 365, Canvas, Azure, and Confluence with 98% first-contact resolution, troubleshooting access and platform workflows.
- Wrote 200+ technical guides for recurring issues, cutting repeat requests 30% and raising satisfaction 20%.

PROJECTS

Fault-Tolerant Payments Ledger | Go, PostgreSQL, Kafka, Docker

- Built a double-entry payments ledger with 0 balance errors across 50,000+ seeded fault-injection tests.
- Prevented duplicate charges on network timeouts using idempotency keys and a transactional outbox to Kafka.
- Sustained 3,200 transfers/sec at <35 ms p99 in load tests, using ID-ordered row locks to prevent deadlocks.

Real-Time Fraud Decisioning Engine | Python, FastAPI, Kafka, Redis, LightGBM, SHAP

- Cut expected fraud losses 36% vs. a rule-based baseline by tuning LightGBM thresholds to business costs.
- Scored 1,500+ Kafka transaction events/sec at <18 ms p99 latency, with real-time velocity features in Redis.
- Attached top-3 SHAP reason codes to every blocked transaction and monitored model drift with PSI.

LLM Cost & Compliance Gateway | Python, FastAPI, Redis, OpenAI API, Sentence-Transformers

- Cut LLM API costs 54% with 94% quality agreement on a 1,000-prompt benchmark via model routing and caching.
- Answered repeat and similar prompts from a Redis semantic cache in <12 ms, skipping redundant LLM calls.
- Redacted personal data (PII) with 96% recall before prompts reach the model; added token-bucket rate limits.

TECHNICAL SKILLS

Languages: Python, Go, C++, C, Java, SQL, JavaScript, TypeScript

Systems & Tools: Linux/Unix, Git, Docker, PostgreSQL, Redis, Kafka, REST APIs, FastAPI, CI/CD

AI, Data & Cloud: Databricks, Azure, Azure AI Foundry, TensorFlow, LightGBM, Flask, Streamlit, Power BI, React

LEADERSHIP & CREDENTIALS

SASE | Workshops Lead (previously Vice President, External): As VP, led 4+ committees, 30+ members, and 10+ partnerships; mentored 60+ students through technical workshops.

HackaBull | Web Development Lead: Scaled event website delivery from solo development to a 3-developer team supporting 500+ registrants.

Honors & Groups: SWE Scholar; 2× Dean's List; 2× Hackathon Winner; Rewriting the Code; Adobe Ambassador.

Certifications: Databricks Certified; NVIDIA DLI Generative AI; Microsoft Azure AI-900 & AZ-900.